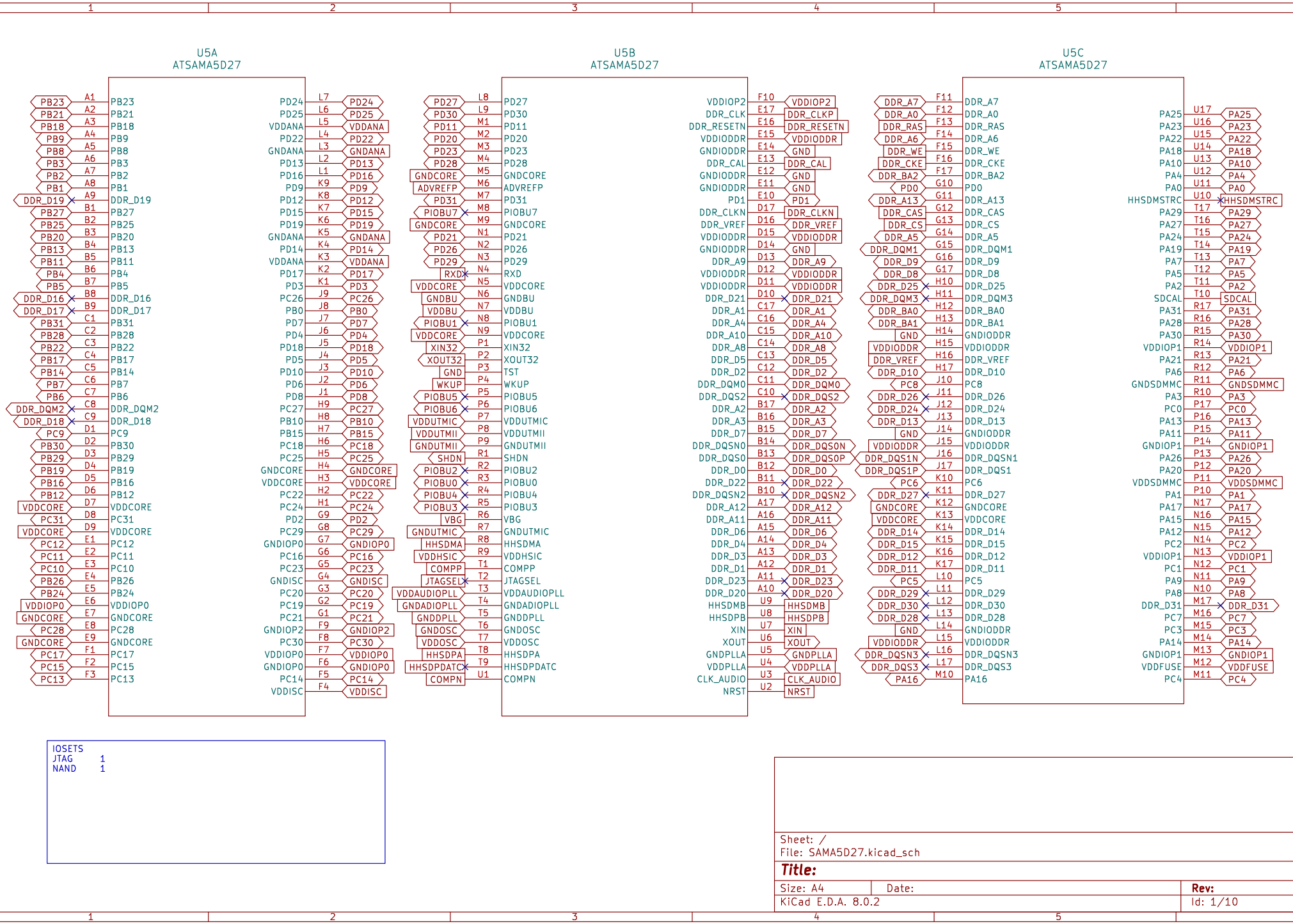
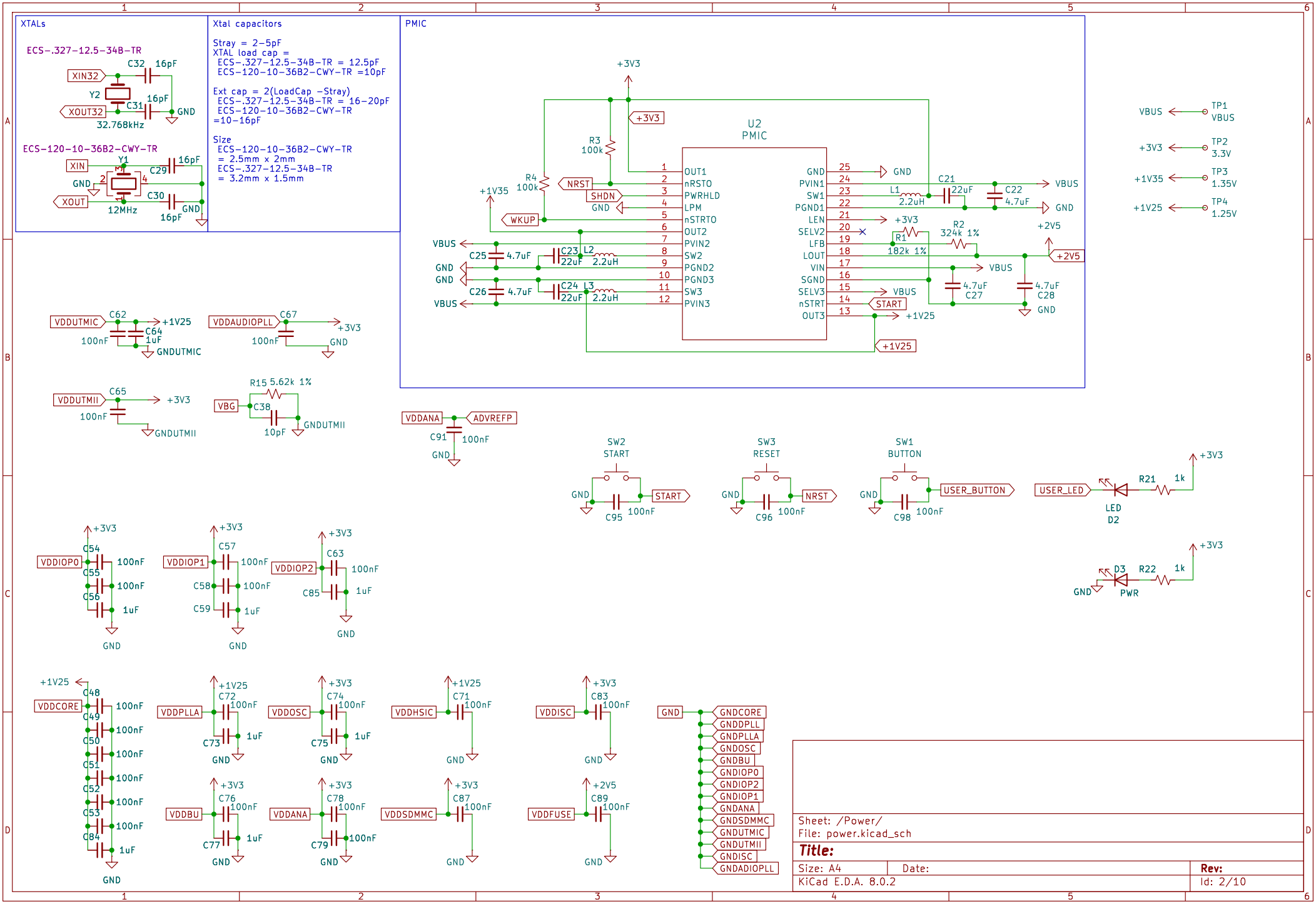
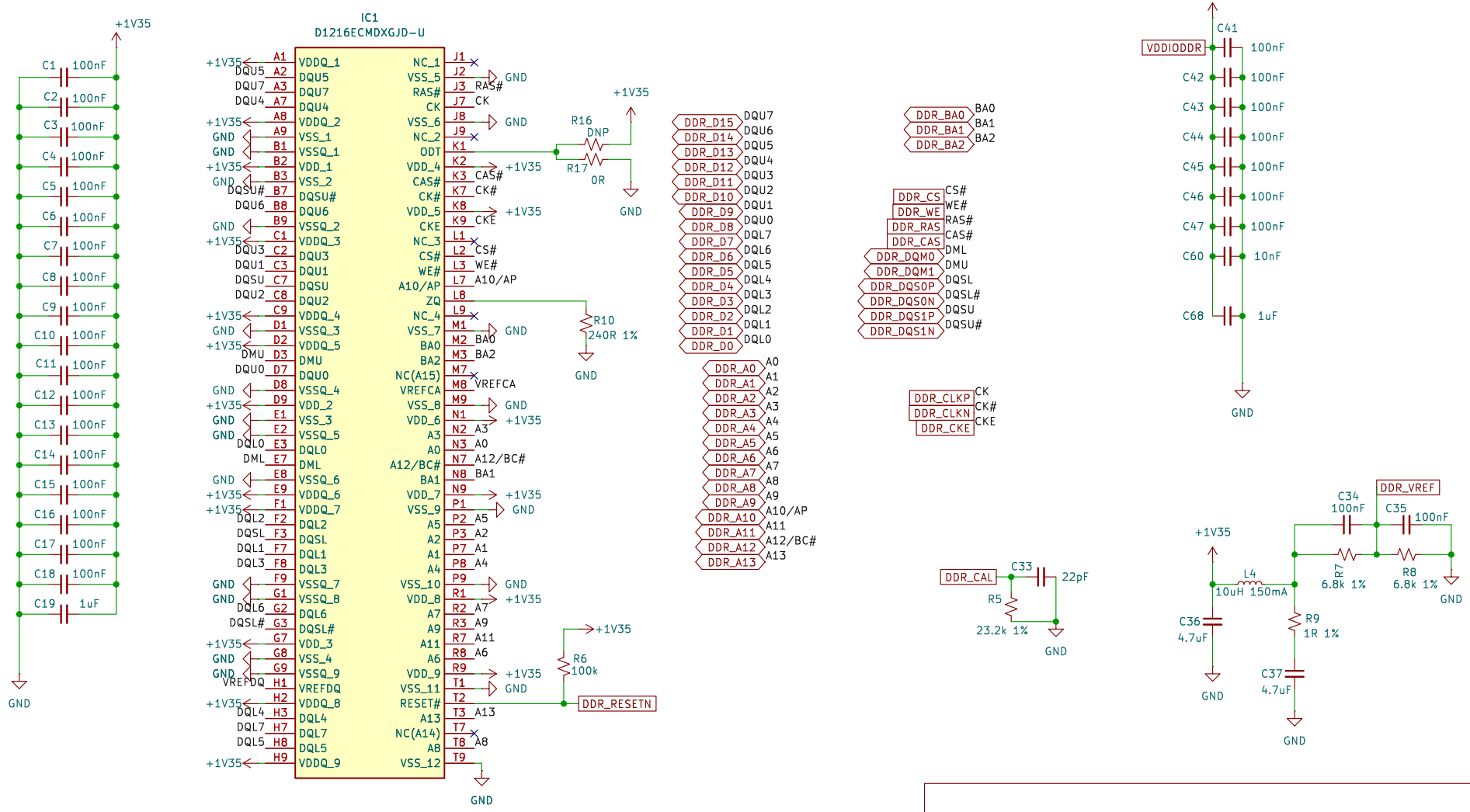

U5A
ATSAMA5D27
U5B
ATSAMA5D27
U5C
ATSAMA5D27
IOSETS
JTAG 1
NAND 1
Sheet: /
File: SAMA5D27.kicad_sch
Title:
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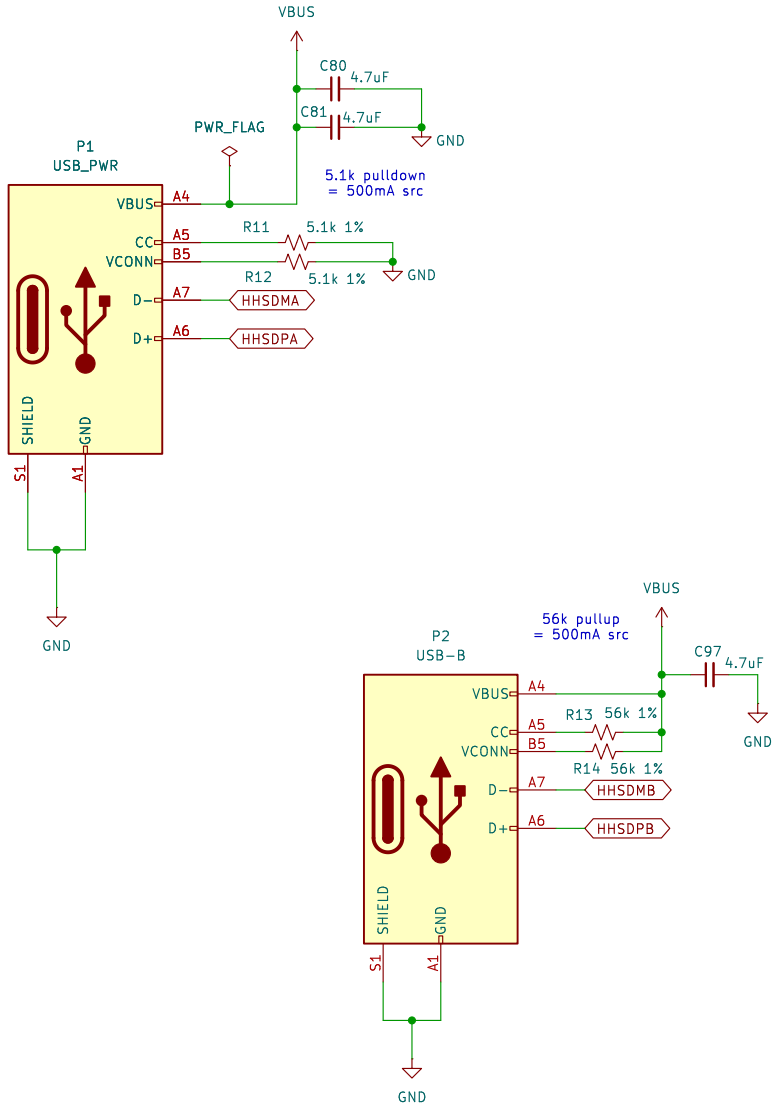

XTALs
ECS-.327-12.5-34B-TR
XIN32
C32 16pF
Y2
C31 16pF
XOUT32
GND
32.768kHz
ECS-120-10-36B2-CWY-TR
XIN
Y1
C29 16pF
GND
2
4
C30
XOUT
12MHz
GND
16pF
Xtal capacitors
Stray = 2-5pF
XTAL load cap =
ECS-.327-12.5-34B-TR = 12.5pF
ECS-120-10-36B2-CWY-TR =10pF
Ext cap = 2(LoadCap -Stray)
ECS-.327-12.5-34B-TR = 16-20pF
ECS-120-10-36B2-CWY-TR
=10-16pF
Size
ECS-120-10-36B2-CWY-TR
= 2.5mm x 2mm
ECS-.327-12.5-34B-TR
= 3.2mm x 1.5mm
PMIC
+3V3
U2
PMIC
R3
100k
R4
100k
NRST
SHDN
GND
WKUP
+1V35
VBUS
C25 4.7uF
C23 22uF
L2 2.2uH
GND
GND
C26 4.7uF
C24 22uF
L3 2.2uH
VBUS
OUT1
nRSTO
PWRHLD
LPM
nSTRTO
OUT2
PVIN2
SW2
PGND2
PGND3
SW3
PVIN3
GND
PVIN1
SW1
PGND1
LEN
SELV2
LFB
LOUT
VIN
SGND
SELV3
nSTRT
OUT3
L1 2.2uH
C21 22uF
C22 4.7uF
R1
182k 1%
R2
324k 1%
+2V5
4.7uF
C27
4.7uF
C28
START
+1V25
VBUS
GND
TP1
VBUS
TP2
3.3V
TP3
1.35V
TP4
1.25V
VDDUTMIC
C62
+1V25
C64
1uF
100nF
GNDUTMIC
VDDAUDIOPLL
C67
+3V3
100nF
GND
VDDUTMII
C65
+3V3
100nF
GNDUTMII
R15 5.62k 1%
VBG
C38
10pF
GNDUTMII
VDDANA
ADVREFP
C91
100nF
GND
SW2
START
GND
START
C95 100nF
SW3
RESET
GND
NRST
C96 100nF
SW1
BUTTON
GND
USER_BUTTON
C98 100nF
USER_LED
R21 1k
+3V3
LED
D2
D3
R22 1k
PWR
VDDIOP0
C54
C55
C56
100nF
100nF
1uF
GND
VDDIOP1
C57
C58
C59
100nF
100nF
1uF
GND
VDDIOP2
C63
C85
100nF
1uF
GND
VDDCORE
C48
C49
C50
C51
C52
C53
C84
100nF
100nF
100nF
100nF
100nF
100nF
1uF
GND
VDDPLLA
C72
C73
100nF
1uF
GND
VDDOSC
C74
C75
100nF
1uF
GND
VDDHSIC
C71
100nF
GND
VDDISC
C83
100nF
GND
VDDBU
C76
C77
100nF
1uF
GND
VDDANA
C78
C79
100nF
100nF
GND
VDDSDMMC
C87
100nF
GND
VDDFUSE
+2V5
C89
100nF
GND
GND
GNDCORE
GNDDPLL
GNDPLLA
GNDOSC
GNDBU
GNDIOP0
GNDIOP2
GNDIOP1
GNDANA
GNDSDMMC
GNDUTMIC
GNDUTMII
GNDISC
GNDADIOPLL
Sheet: /Power/
File: power.kicad_sch
Title:
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Sheet: /DDR/
File: ddr.kicad_sch

Title:

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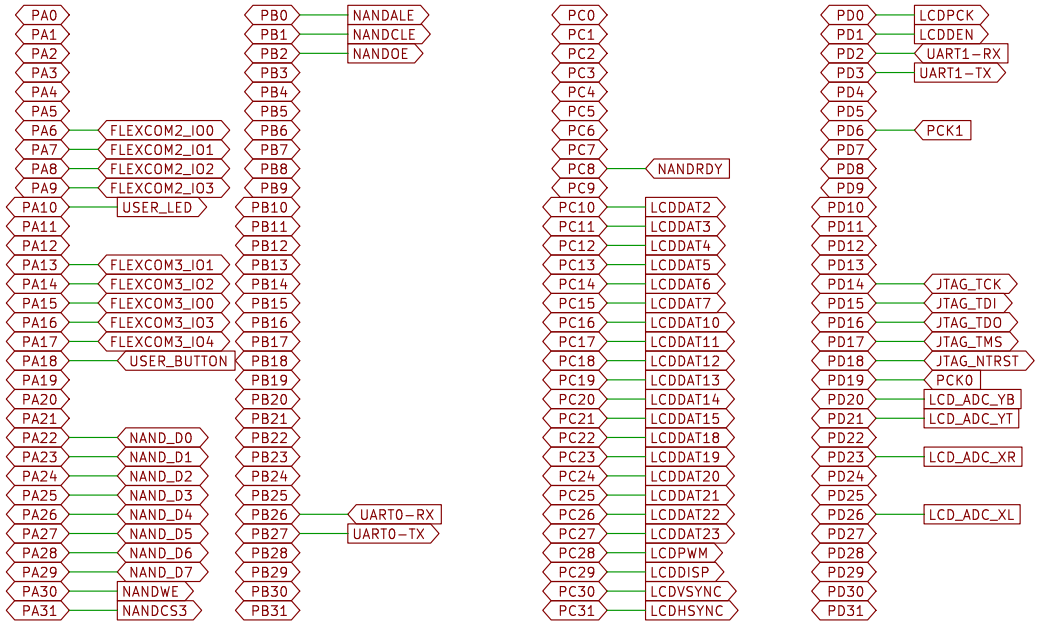
USB−A can be device
(default) or host
USB−B is host

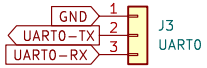
USB CC pins

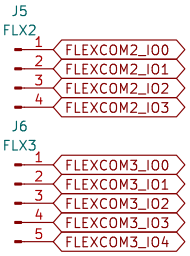
For UFP:
5.1k (10%) pulled down asks
for 1.5A @ 5V

For DFP:
56k (20%) pulled up offers
500mA with USB2.0

Sheet: /USB/		
File: usb.kicad_sch		
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PIO	IOSET
NAND	1
LCD	2
JTAG	1
FLEXCOM2	1
FLEXCOM3	1
PCK0	1

Check for NC
These are unused

RXD (serial wakeup pin)
DDR-D15-D31
DDR_DQS2 DDR_DQS3
DDR_DQM2 DDR_DQM3
DDR_DQSN2 DDR_DQSN3
HHSDPDATC HHSDMSTRC (HSIC)
PIOBU0-7

FLEXCOM2

FlxPins	Pins	UART	SPI
IO0	PA6	TXD	MOSI
IO1	PA7	RXD	MISO
IO2	PA8	SCK	SPCK
IO3	PA9	CTS	NPCS0

FLEXCOM3

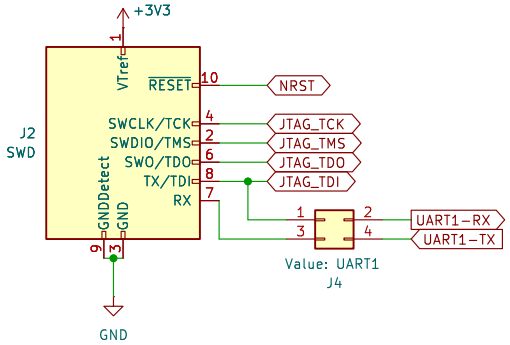
FlxPins	Pins	UART	FLEX_SPI	BOOT_SPI0
IO0	PA15	TXD	MOSI	MOSI
IO1	PA13	RXD	MISO	–
IO2	PA14	SCK	SPCK	SPCK
IO3	PA16	CTS	NPCS0	MISO
IO4	PA17	DTR xx	NPCS1	NPCS0

Confusing: FLEX_SPI is Flexcom in SPI mode, but SPI0 is another peripheral MUXd with Flexcom
The boot code picks this SPI0 to boot from
Slightly different pinout between the two SPIs as well

Sheet: /PIO/
File: pio.kicad_sch

Title:

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JTAG uses JTAG IOSET1
This affects Flexcom4 and
other minor pins

JTAG 20pin pinout

				SWD+VCOM
VTREF	1	2	NC	VTREF
nTRST	3	4	GND	NC
TDI	5	6	GND	(5-TX)
TMS	7	8	GND	(7-SWDIO)
TCK	9	10	GND	(9-SWCLK)
RTCK	11	12	GND	NC
TDO	13	14	GND	(13-SWO)
nRST	15	16	GND	nRST
NC	17	18	GND	(17-RX)
5V	19	20	GND	5V

JTAG 10pin pinout

VCC	1	2	SWDIO/TMS
GND	3	4	SWDCLK/TCK
GND	5	6	SWO/TDO
KEY	7	8	NC/TDI(TX)
GNDDetect		10	nRST

SAMA5D2 has no RTCK, eval has it NC

Sheet: /JTAG/
File: jtag.kicad_sch

Title:

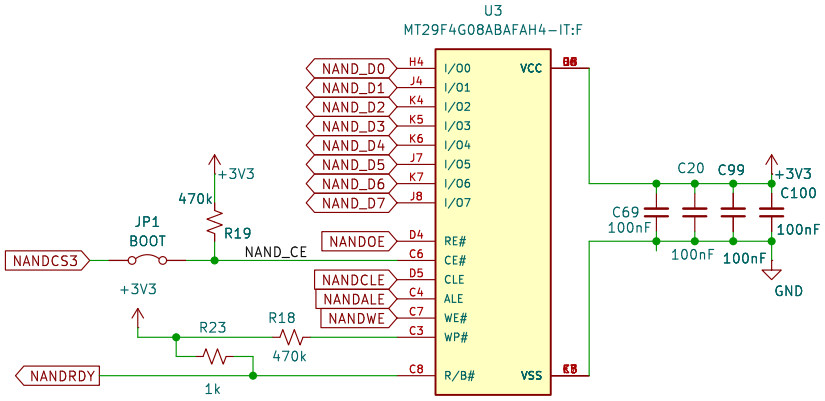
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Sheet: /AUDIO/
File: audio.kicad_sch

Title:

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NAND uses NAND IOSET1



Sheet: /NAND/
File: nand.kicad_sch

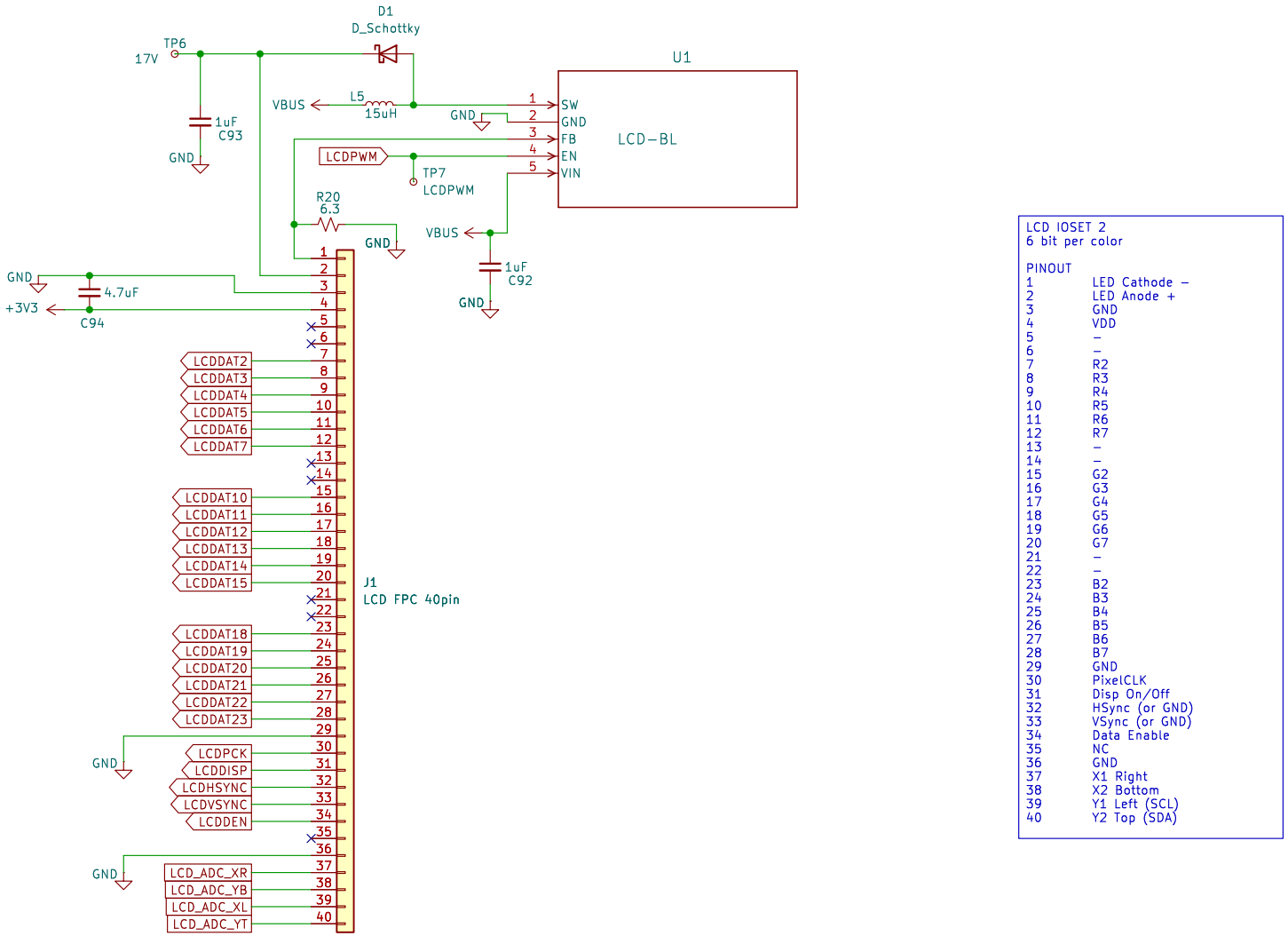
Title:

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Sheet: /QSPI/
File: qspi.kicad_sch

Title:

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LCD IOSET 2
6 bit per color

PINOUT

1	LED Cathode –
2	LED Anode +
3	GND
4	VDD
5	–
6	–
7	R2
8	R3
9	R4
10	R5
11	R6
12	R7
13	–
14	–
15	G2
16	G3
17	G4
18	G5
19	G6
20	G7
21	–
22	–
23	B2
24	B3
25	B4
26	B5
27	B6
28	B7
29	GND
30	PixelCLK
31	Disp On/Off
32	HSync (or GND)
33	VSync (or GND)
34	Data Enable
35	NC
36	GND
37	X1 Right
38	X2 Bottom
39	Y1 Left (SCL)
40	Y2 Top (SDA)

Sheet: /LCD/
File: lcd.kicad_sch

Title:

Size: A4	Date:	**Rev:**
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